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ME3330 – Filters Technical Summary

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Deliverable 1:

This technical summary is an overview of the creation and testing of a high pass filter. The filter is tested at different frequencies, and the gain is measured in decibels. This can be seen plotted against frequency below in FIG. 1.

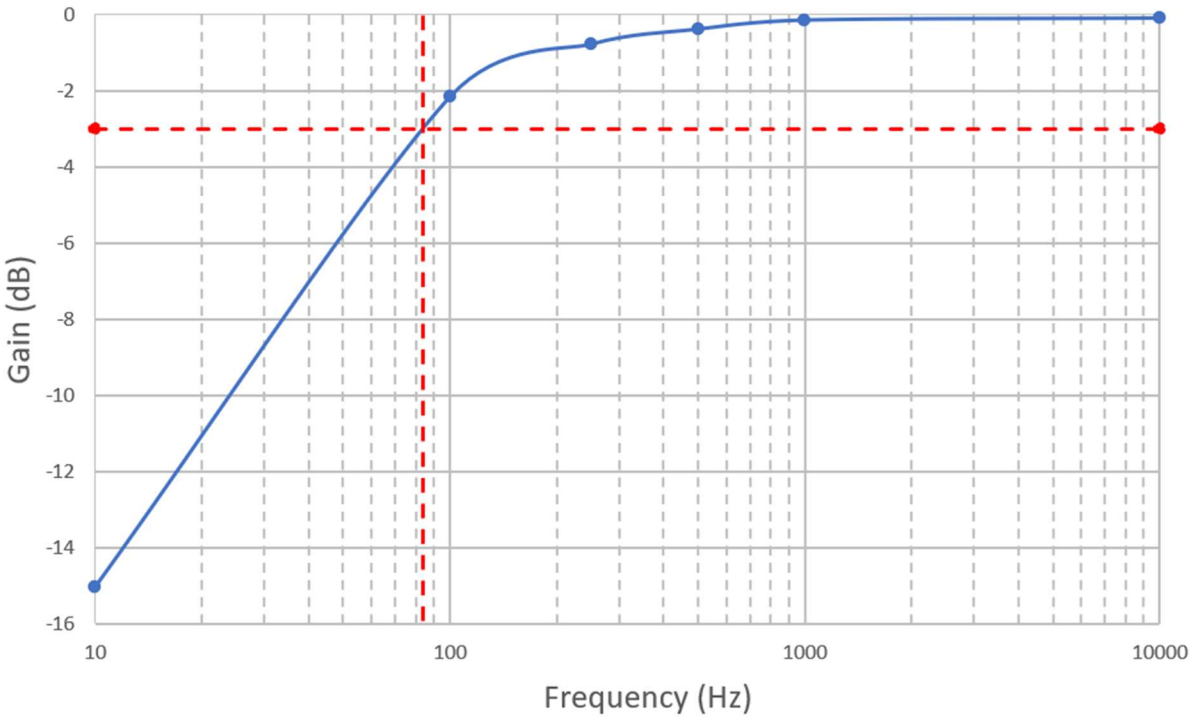


FIGURE 1. Experimental data plotted on bode plot, intersection shown at the red lines. The intersection occurs at approximately 84 Hz

The experimental cutoff frequency was found by finding the intersect at -3 dB. The values plotted are listed in TABLE 1 below.

Input Frequency (Hz)	Input Voltage (V)	Output Voltage (V)	Gain	Gain (dB)
10	0.502	0.089	0.177290837	-15.02627421
100	0.500	0.390	0.78	-2.158107946
250	0.492	0.450	0.914634146	-0.77505178
500	0.491	0.470	0.957230143	-0.379672684
1000	0.490	0.482	0.983673469	-0.142980836
10000	0.490	0.485	0.989795918	-0.089086829

TABLE 1: This table contains experimental values. The input frequency stepped up as the input voltage and output voltage were recorded for all frequencies. This is necessary to calculate the gain, which is then converted from being unitless to dB using equation 1.

$$G_{dB} = 20\log_{10}(\frac{V_o}{V_l}) \tag{1}$$

Deliverable 2:

A high pass filter is constructed differently from a low pass filter. The position of the capacitor(s) and resistor(s) are switched. This leads to a different transfer function. The derivation of the transfer function can be seen below in Equation 2.

$$\frac{V_o}{V_i} = \frac{Z_R}{Z_R + Z_C} = \frac{R}{R + \frac{1}{sC}} = \frac{sRC}{1 + sRC} = \frac{j\omega RC}{1 + j\omega RC} = \frac{2\pi f_c RC}{1 + 2\pi f_c RC} \tag{2}$$

Now that the transfer function has been derived, the function can be further solved for the gain by taking the absolute value. This will then be converted to dB in Equation 3.

$$G_{dB} = 20\log_{10}(|\frac{V_0}{V_i}|) = 20\log_{10}(\frac{2\pi f_c RC}{\sqrt{1 + (2\pi f_c RC)^2}}) \tag{3}$$

This transfer function was solved for a high pass filter, and the circuit diagram for the high pass filter is displayed below in FIG. 2.

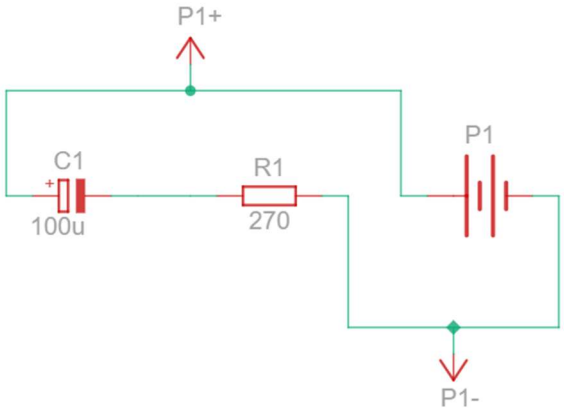


FIGURE 2: Circuit diagram for a generic high pass filter.

Equation 2 was solved using MATLAB, and the output at -3 dB can be seen below in FIG. 3. This value of 58.9 Hz does not necessarily mirror the experimental results, but this will be discussed later in deliverable 3.

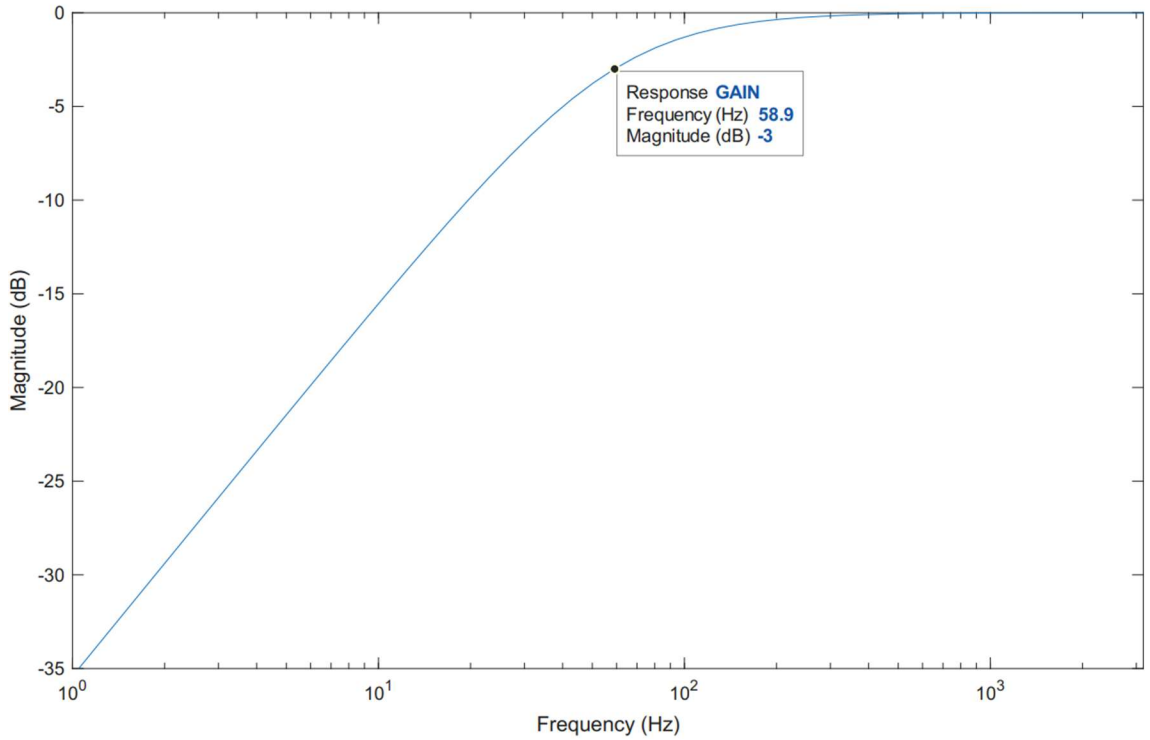


FIGURE 3. Ideal Bode Diagram from MATLAB with datapoint at the -3dB where the cutoff frequency occurs.

The transfer function has been solved above in Equation 2, and can be seen experimentally in FIG. 1, therefore now the cut-off frequency will be calculated below. The exact cutoff frequency occurs when the power transfer is at ½ of the maximum value. Using the gain equation, Equation 1, and the power transfer maximum value, the equation can be rearranged to solve for cutoff frequency.

$$f_c = \frac{1}{2\pi RC} \tag{4}$$

R is the resistance in ohms, and C is the capacitance of the capacitor. Plugging in the values from the circuit, a theoretical cutoff frequency of 58.95 Hz is found. This value is reflected in FIG. 3, but not in FIG. 1.

Deliverable 3:

The theoretical cut-off frequency values calculated were 58.95 Hz, and the experimental cut-off frequency was 84 Hz. This is a large gap between these values, and this could be due to a combination errors.

Major Talking Point (1):

The value of the capacitance was not directly measured, and due to its location at the bottom of Equation

2, a smaller capacitance value would yield a larger cut-off frequency. With a tolerance of 20% in the capacitance, this will yield a new potential cut-off frequency of 73.68 Hz. This is possible due to lower quality capacitors having tolerances that can be as wide as 20%.

Major Talking Point (2):

The other potential source of error is a lack of data points. There are only 5 values where recorded, and the possibility of error is large in the approximation of the line. This is exacerbated by the fact that the plot seen in FIG. 1 is also on a logarithmic scale. More data points have the possibility to greatly increase the accuracy of the readings. This can be further inferred by the difference in the shapes of FIG. 1 and FIG. 3. The cut-off frequency was taken at 3dB at both, but one of the sections is linear in FIG. 1, and one is curved in FIG. 3.

These errors could combine to create the disparity between the experimental and theoretical cut-off frequencies.

Deliverable 4:

In order to design a high-pass filter, begin with the capacitor(s) and follow with the resistor. They should be wired in series. As discussed previously, to increase the cutoff frequency, a lower capacitance is required. The solution I discovered is to wire 3 capacitors in series. This has the combined impendence shown in Equation 5 below.

$$C = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2} + \ldots} \tag{5}$$

The values of these capacitors are 33 µF, 10 µF, 6.8 µF, and 6.8 µF. These are all using standard values for capacitors. The resulting value when plugged into Equation 2 is 250.18 Hz. The circuit diagram can be seen below in FIG. 4.

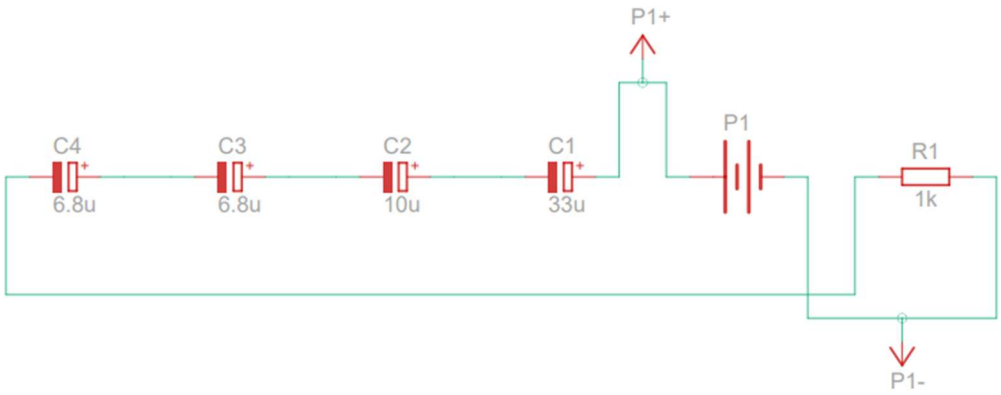


FIGURE 4: Circuit diagram for the 250Hz high pass filter using a 270Ω Resistor.

REFERENCES

[1] Department of Mechanical Engineering, Clemson University. *ME 3330 – Mechanical Engineering Lab II Student Manual.* Fall 2025 ed., Clemson University, 2025.

[2] *Electronics Tutorials*, "Capacitor Characteristics and Specifications," Electronics-Tutorials.ws. [Online]. Available: http://electronics-tutorials.ws/capacitor/cap_3.html. [Accessed: Jan. 21, 2026].